CAPSTONE PROJECT SUBMISSION TEMPLATE

Template 6: Performance Testing

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 6: Performance Testing**
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GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. Test 1: Data Volume Ingested Across Architecture Layers

Entity / Dataset	RAW Landing Rows	STAGING Cleansed Rows	CURATED Model Grain	Validation Status
Customers	1,500	1,500	1,500 (DIM_CUSTOMERS)	**100% Match**
Restaurants	80	80	80 (DIM_RESTAURANTS)	**100% Match**
Menu Items	612	612	612 (DIM_MENU_ITEMS)	**100% Match**
Orders	12,000	12,000	12,000 (FACT_ORDERS)	**100% Match**
Order Line Items	32,921	32,921	32,921 (FACT_ORDER_ITEMS)	**100% Match**
Total Record Footprint	**47,113 Rows**	**47,113 Rows**	**47,113 Total Records**	**Zero Records Dropped**

2. Test 2: Filter Efficiency & Micro-Partition Pruning Benchmark

Query Execution Scenario	Partitions Total	Partitions Scanned	Pruning Rate	Execution Latency	Speedup Factor
Full Table Scan (Baseline)	12	12 (100%)	0.0%	312 ms	1.0x (Baseline)
Date-Pruned Query (Oct 2024)	12	1 (8.3%)	**91.7% Pruned**	**48 ms**	**6.5x Faster**
Snowflake Result Cache Hit	12	0 (Cached)	100% Cached	**12 ms**	**26.0x Faster**

3. Test 3: Audit of Engineered Analytical Fields (24 New Columns)

A total of **24 distinct analytical columns** were engineered during the ETL pipeline across Staging and Curated layers:

- **Customer Dimension (4):** customer_key (MD5 hash), loyalty_segment (VIP/Frequent/Occasional/Single), is_repeat_customer (Boolean), customer_aov (Lifetime spend / orders).
- **Restaurant Dimension (2):** restaurant_key (MD5 hash), rating_tier (Top/High/Avg/Low), cost_tier (Budget/Mid/Premium).
- **Menu Item Dimension (2):** item_key (MD5 hash), price_tier (Budget/Mid/Premium).
- **Time Dimension (2):** time_key (Surrogate hash), meal_period (Breakfast, Lunch Peak, Snacks, Dinner Peak, Late Night).
- **Orders Fact (7):** order_key, date_key, realized_revenue, basket_item_quantity, is_delivered, is_delivered_late, is_cancelled.
- **Order Items Fact (2):** order_item_key, item_total_price (quantity * unit_price).
- **Audit Lineage (5):** stg_loaded_at, curated_loaded_at, is_realized_flag, discount_percentage, fulfillment_sla_mins.

4. Test 4: Snowflake Query History & Resource Profiling

Audited via TABLE(INFORMATION_SCHEMA.QUERY_HISTORY()): Average query latency across all dashboard operations remained at **84 ms**. The virtual warehouse auto-suspended within 60 seconds of idle time, consuming fewer than 0.15 Snowflake credits for the entire testing session.